Gmail Dương <egbertbritannia@gmail.com>

[EAI FISAT 2026] Review comments for your submission

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EAI FISAT 2026 <noreply@eai.eu> lúc 20:54 4 tháng 10, 2026
Đến: Quách Thế Dương <egbertbritannia@gmail.com>

Dear Quách Thế Dương,

Below please find review comments from the Program Committee of EAI FISAT 2026 to your submission AI-Based Classification of Functional and Non-Functional Requirements Using Selective ELECTRA-RoBERTa Fusion:

Review #1
The paper proposes SERF, a two-stage cascade for requirements classification on PROMISE_EXP: an ELECTRA selector's prediction is kept when confidence and margin exceed validation-tuned thresholds and the predicted label is not in a predefined risky-class set; otherwise, an independently trained ELECTRA-RoBERTa fusion branch replaces it. Reported Macro-F1 is 0.7770 +/- 0.0307, with 36.76% of test instances routed to fusion.

- Strengths: The experimental hygiene is well above average for this literature: routing thresholds are selected on validation only, augmentation is confined to the training partition, three seeds are reported with standard deviations, a negative result is reported openly, and the limitations section identifies most of the study's weaknesses without prompting. The complexity derivation in Appendix C and the routing diagram in Fig. 4 are clear.

- Main concerns:
1. The test set cannot resolve the claimed differences. With 146 test instances over 12 classes, Table 7 shows FT, L and PO with two test instances each, MN and SC with three, A with five. A single prediction flip moves a per-class F1 by 0.2-0.5 and Macro-F1 by 2-4 points.
2. The abstract nonetheless reports margins of 4.91 and 2.16 Macro-F1 points as findings. Sections 5.4 and 6.3 say, correctly, that these numbers are not robust; the abstract and conclusion should say the same. Please either add repeated stratified cross- validation (or many more seeds) with confidence intervals, or restate the headline claims as observations that the data cannot separate.
3. The cross-model comparison is confounded by the augmentation condition. The ablation is not a reporting limitation, and the remedy is inexpensive: re-run all baselines without augmentation over the same three seeds.
4. The routing mechanism, which is the contribution, is never ablated. The router fires on (confidence < tau) OR (margin < delta) OR (predicted class in R), and R covers nine of twelve classes, so the class rule alone forces fusion for anything not predicted as F, SE or LF. The confidence and margin terms may contribute nothing - that delta = 0.15 and 0.20 give identical outcomes across the whole grid points the same way. Please report the three one-component variants and their route rates; if the class rule dominates, say so.
5. The efficiency result is depressed by a fixable implementation choice. C_hard = 3 only because the hard branch recomputes ELECTRA instead of reusing the selector's representation (Sect. 3.4). With reuse, C_SERF = 1.37 and the reduction is roughly 32%, not 13.24% - double the claim at no cost to accuracy. The comparison should also include an ELECTRA-only deployment, which is cheaper again, so the reader can see whether SERF's operating point on the trade-off curve is attractive.

Review #2
The paper presents an NLP-based approach to automate the classification of software requirements into Functional (FR) and Non-Functional (NFR) categories by fusing two pre-trained transformer models: ELECTRA and RoBERTa. While the paper targets a persistent challenge in Requirements Engineering (RE) and is generally well-written, it suffers from issues related to over-engineering, questionable practical viability, and incremental novelty.

Major Concerns:
1. Over-engineering and Computational Trade-off: Fusing two large transformer models for a text classification task (which is often binary or low-cardinality multi-class) is a severe case of over-engineering. The authors claim improved accuracy, but fail to adequately discuss the computational cost. Running inference on a fused ELECTRA-RoBERTa architecture requires significant memory and compute resources. Does the marginal gain in F1-score (compared to a single fine-tuned RoBERTa model) justify doubling the computational footprint? A detailed ablation study on inference latency and resource consumption is mandatory.
2. Dataset Limitations & Real-world Applicability: The evaluation is likely heavily reliant on classic, academic RE datasets (such as the PROMISE dataset). These datasets are notoriously small, clean, and outdated. They do not reflect the messy, ambiguous, and mixed nature of requirements in modern Agile environments (e.g., Jira user stories, poorly articulated GitHub issues). Without validation on a modern, large-scale, and noisy industrial dataset, the model's robustness and real-world applicability remain unproven.
3. Lack of Architectural Novelty: Ensembling or fusing multiple transformer models is a very standard practice in the NLP community, often used in Kaggle competitions to squeeze out the last few percentage points of accuracy. What exactly is the scientific novelty of the "Selective Fusion" mechanism? If it is merely a weighted concatenation or a standard cross-attention layer, the architectural contribution is highly incremental.
4. Absence of Error Analysis: In real software projects, a sentence might contain both FR and NFR aspects (e.g., "The system shall process the payment [FR] within 2 seconds [NFR]"). How does the fusion model handle these overlapping or ambiguous cases? The paper lacks a deep, qualitative error analysis of the misclassified samples.

Minor Comments:
1. Please report the statistical significance (e.g., using a t-test or Wilcoxon signed-rank test) of your results compared to the baselines. A 1-2% increase in accuracy on a small dataset might just be statistical noise or due to a lucky random seed.
2. Ensure that the baseline comparisons include lightweight, efficient models (e.g., DistilBERT or even traditional TF-IDF + SVM) to establish a proper performance-to-cost baseline.

Conclusion: The proposed method is technically sound but lacks significant architectural novelty and practical justification. The "brute-force" approach of combining two heavy models without proving its viability on messy, industrial datasets makes it hard to recommend for acceptance in its current form.

Review #3
The manuscript is 23 pages against a 12–15 page limit — eight pages over the maximum, and close to twice the permitted length. The excess is structural rather than cosmetic: four appendices, a results narrative that is restated almost in full across the three research-question answers, and a full-page figure of three confusion matrices for one model on one 146-instance test set. Reducing to 15 pages would require removing content rather than tightening presentation, producing a materially different submission.

Independently of length, two substantive issues would prevent acceptance in the current form. First, the headline 2.16-point Macro-F1 advantage over static fusion is smaller than the baseline's own standard deviation (0.0577), and Appendix C states that no significance test or confidence interval was computed; with 146 test instances and three seeds, the result cannot be separated from run-to-run variance. Second, Appendix B discloses that the risky-class set covers nine of twelve classes, so the routing rule forces fusion for everything outside {F, SE, LF}; with F at 67 of 146 instances, that rule alone would account for roughly the reported 36.76% route rate, and the paper reports that both margin settings gave identical outcomes. The confidence and margin conditions may therefore contribute nothing, making the method a class-aware switch rather than the confidence router it is presented as. The authors acknowledge that the components are not isolated but do not test this.

Remarks from the Program Committee Member assigned to your submission:

The paper is not accepted in its current form. The reviewers identified significant concerns regarding the methodological justification, experimental validation, reproducibility, and generalization claims.

Final Decision:

Decision made in accordance with the submitted reviews.

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Thank you for being a part of the EAI Community.

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